

Background

- The MATH2703 module is assessed by an examination (80%) and a practical (20%). This is the practical, worth 20% of your final module mark.
- Reports should be no more than five sides of A4 paper in length. Reports must be submitted via Turnitin, please name your report file as StudentID-Name-MATH2703.
- Practical is set in week 5 (Monday 23 February 2026). There is no practical session, but some of the lecture (e.g. week 9) and workshop (e.g. week 8) times will be reserved to answer your questions related to practical.
- You can share ideas with other students, but the work you hand in must be your own.
- Please include a signed Declaration of Academic Integrity Form, you can download one from Minerva → Assessment and Feedback.
- You must hand in your solutions by **Monday 27 April 2026 at 14:00**. Please hand in an electronic version via Minerva → Assessment and Feedback → Submit My Work → Turnitin-Coursework. Late work will be penalised at the rate of 5% of the available marks per calendar day.

Data

The data are taken from a database about slave voyages. Specifically, we will consider the yearly estimated number of slaves who embarked on a voyage across the Atlantic Ocean – on a ship under a British flag.

For this practical you will need to obtain the data set $X = (X_1, \dots, X_n)$ from the file “slavery.txt” which you can download from Minerva → Coursework including dataset. Once you have the file “slavery.txt” saved in the folder where you are running R, use the command `xf <- read.table("slavery.txt")` to read the data into R and name the data `xf`. This will give you a data frame called `xf` which has two components: `xf$year`, and `xf$num`. (Try `summary(xf)` or `str(xf)`.)

The task

The task of this practical is to analyse these data, using the following list of points as a guide. Your analysis should consider the points below, but your report should *not* be laid out as “answer 1”, “answer 2”, Instead, structure your report in whichever way you prefer to make it easy to read.

- Plot the data (the variable `num` we will denote by X) and examine any prominent features. Comment on your findings. Use linear regression to remove any linear trend and/or seasonal effects which you believe to be present in the data. Denote the residuals after removing the trend and/or seasonal effects by Y .
- Inspect the process Y and comment on whether an MA or AR process might be suitable to explain any structure present.

- For each of $p = 1, 2, 3$, using the Yule-Walker equations, fit an $AR(p)$ model to the time series Y . You can either use the R function `ar` or solve the Yule-Walker equations “by hand”.
- For each $p = 1, 2, 3$, consider the residuals of the $AR(p)$ model. Plot a correlogram of the residuals, run Ljung-Box test on the residuals, and comment on how well the three models fit the data.
- For each $p = 1, 2, 3$, re-fit the AR parameters using the command `arima`. Choose the model with the best fit based on the AIC criteria. Provide a summary of the final model you have chosen for the original time series, including all the fitted parameters, and their standard errors.
- Using the command `predict` on the result of the `arima` fit (the best model), obtain a forecast for the next 5 years (i.e. 1808–1812). Interpret and discuss.

Guidance on the report

Don’t forget to look for ideas in the example R code shown in the lectures, workshops and the files we have used. Of course, you may use the help system in R, e.g. `help(predict)`.

You should take some care with the presentation of your results. This includes using a clear structure and layout, careful explanation of your results and how you obtained them, meaningful plots with appropriate labels, *etc.* You should start your report with a short (1 paragraph) summary of your findings, written in a style suitable for a non-statistician.

Note that large amounts of R output are *not* needed. Use R output sparingly or not at all. If you do include R output, make sure it is in a fixed-width font like `this` – Courier in Word or verbatim/texttt in L^AT_EX. You do not need to attach your R script.

Do not repeat large sections of theory from the notes – I already know what is in them! Use your limited page count to describe your analysis and conclusions.

If you are interested in the database, you can find out more about this [here](#). However, a square root transformation has been taken to simplify the analysis.